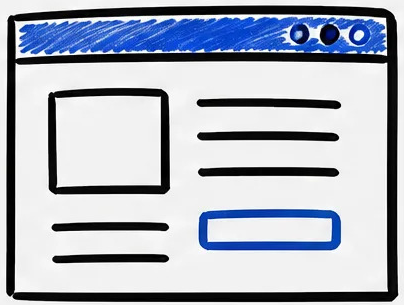
WebSphere v9 Administration: Module 7

Dr. Lee

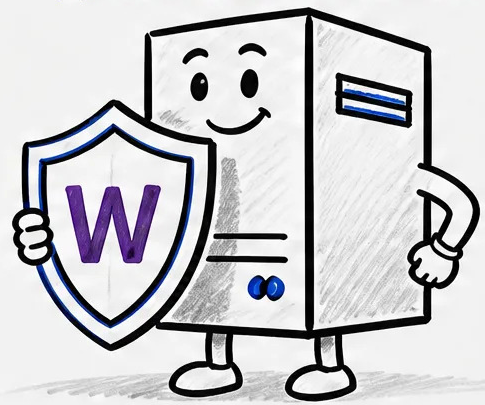
Connect to a Database

APPLICATION



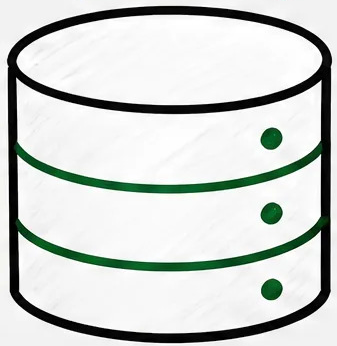
- Sends request
- Receives data

WEBSPHERE v9 APPLICATION SERVER



- Manages Connections
- Provides Security
- Optimizes Resource Use

DATABASE (BACKEND)

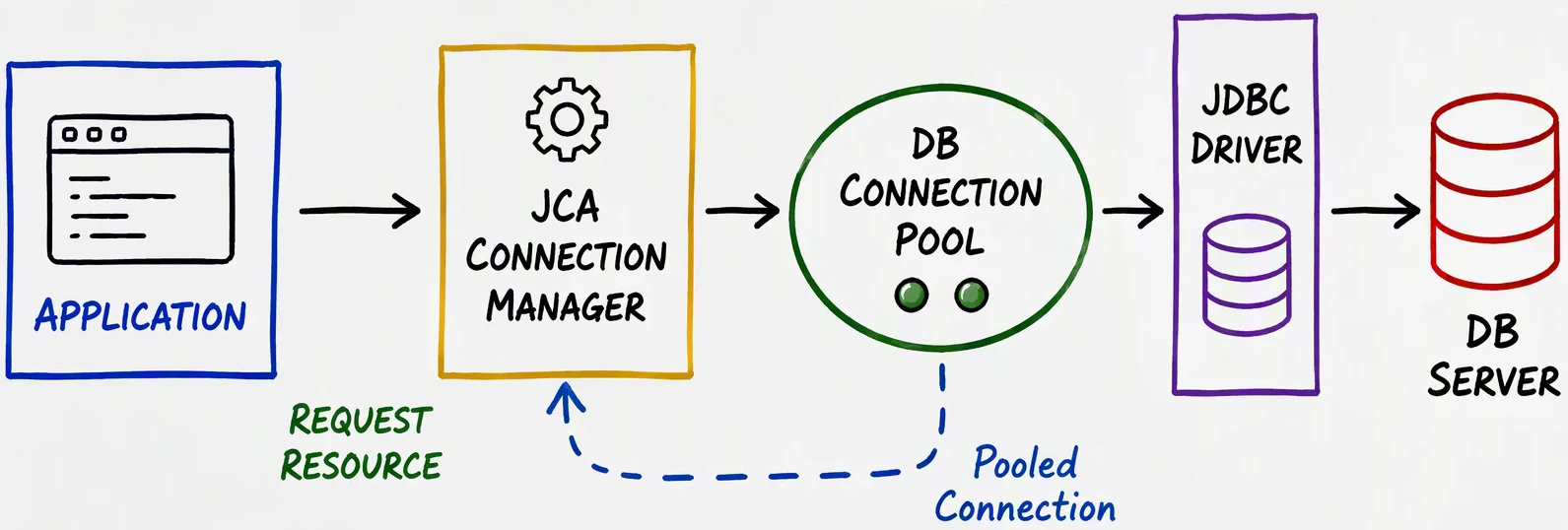


- Stores Data
- Processes Queries



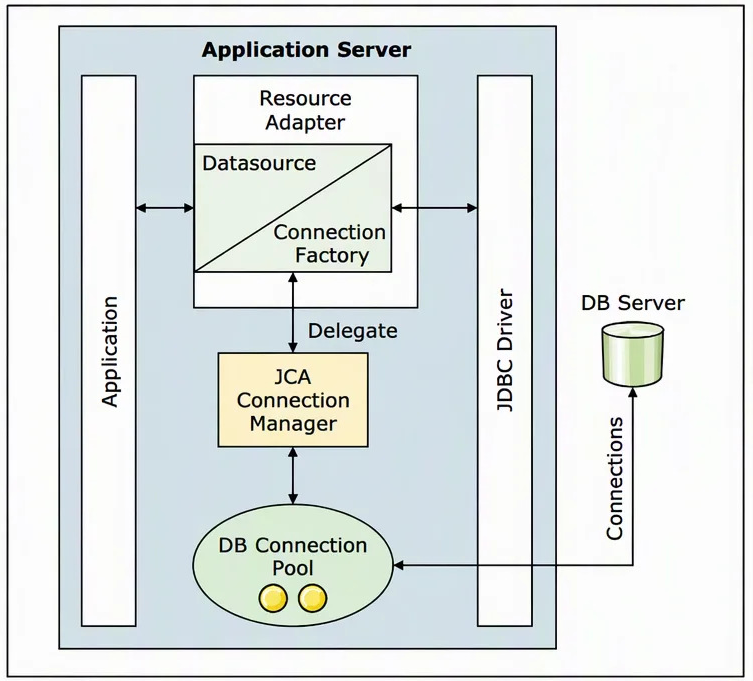
We do not let applications talk to the database directly; we let WebSphere manage the connection.

The Database Access Architecture



The JCA Broker

REFERENCE ARCHITECTURE



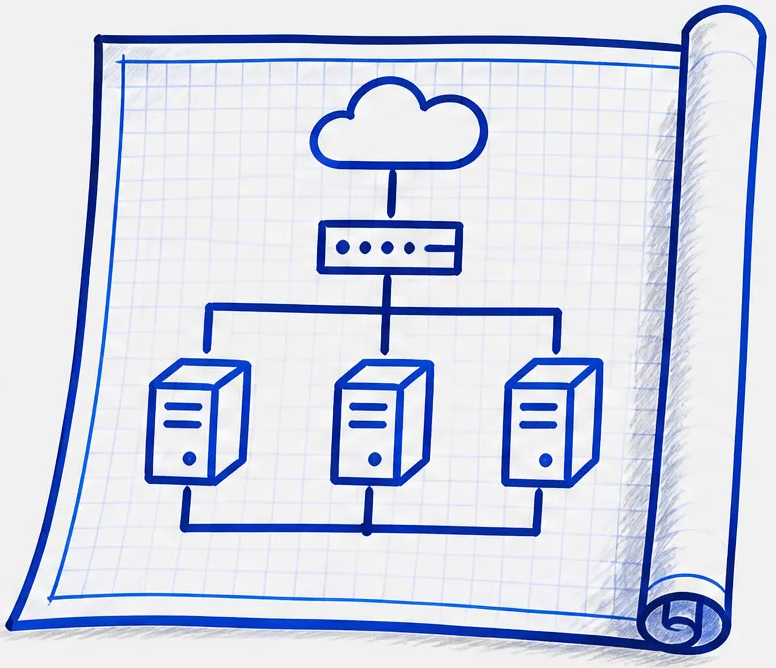
The application asks for a resource; WebSphere provides a pooled connection.

Step 1: J2C Authentication Aliases



Stores and Protects
Security Credentials

Decoupling Security

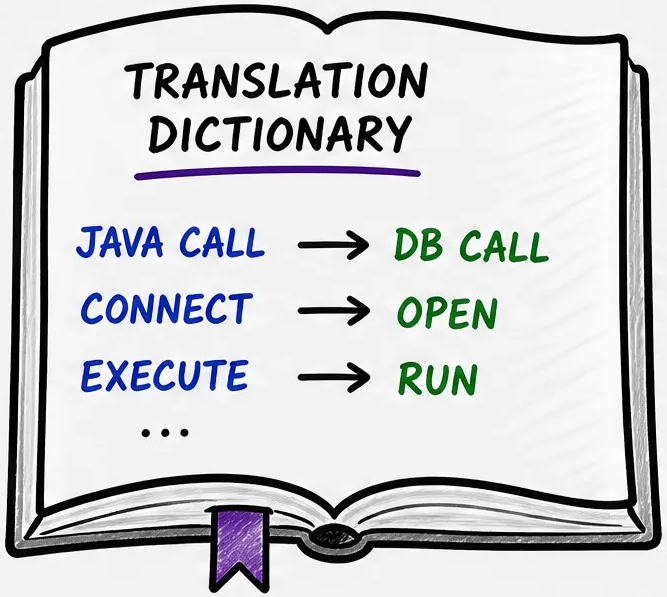


Defines and Manages
Network Configuration

The alias separates security credentials from network configuration.

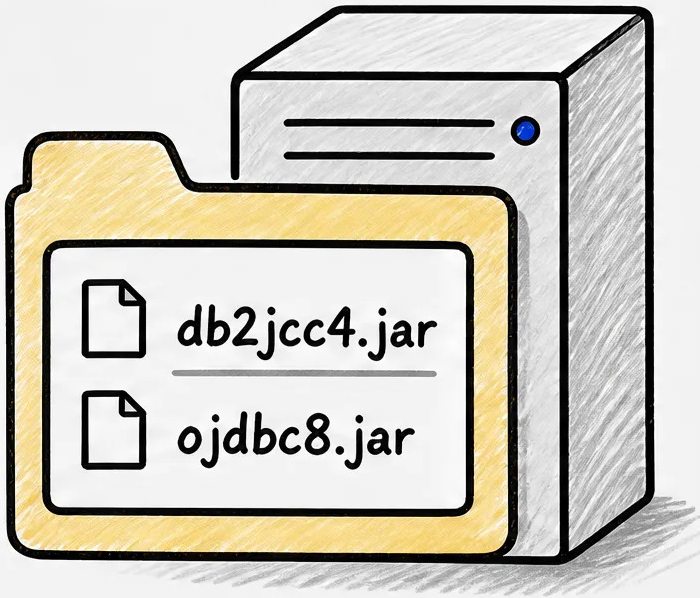
Step 2: The JDBC Provider

JDBC PROVIDER



THE TRANSLATION LAYER

ON SERVER DISK





The JDBC Provider is the translation layer between Java and the database.

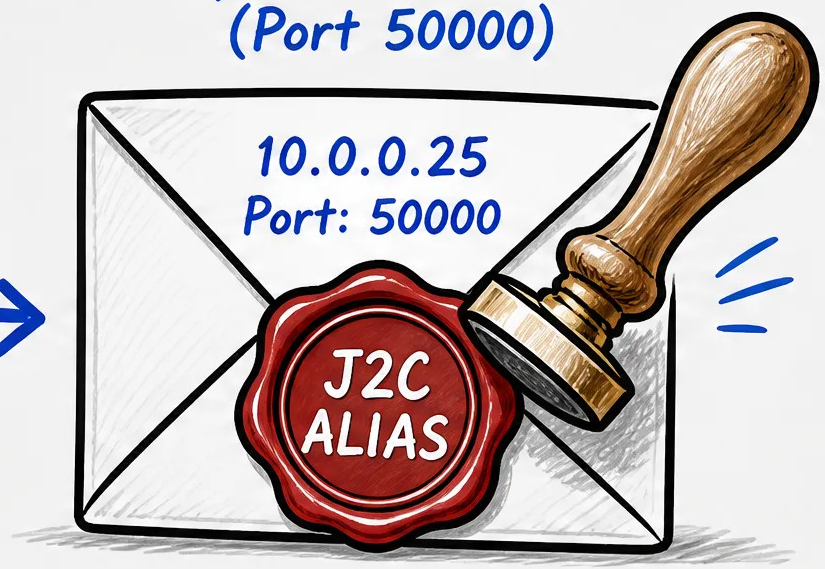
Step 3: The Data Source

JNDI Name



The Destination Address

Specific Address (Port 50000)



Stamped with J2C Alias

The data source is the specific destination address.

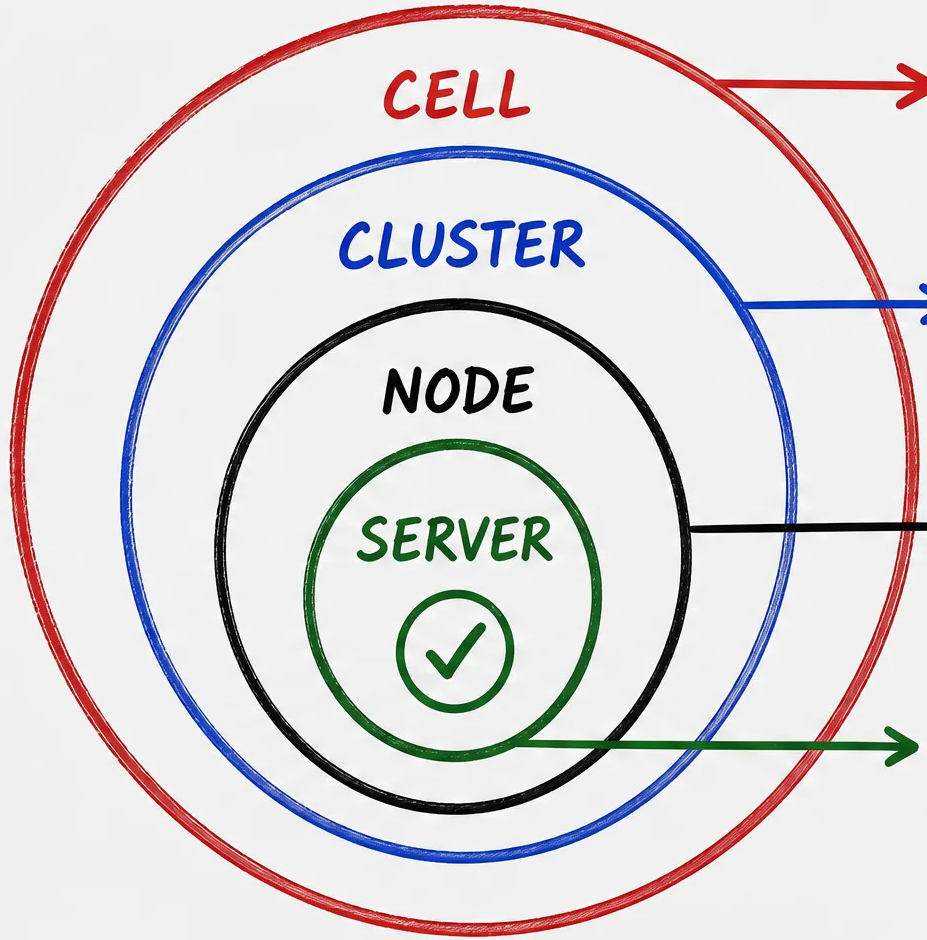
ProTech
protechtraining.com

Data Source Scopes

Dr. Lee

Controlling Visibility

★ Scope controls visibility and impact.



CELL (WIDEST)
Visible across the entire cell.

CLUSTER
Visible within the cluster.

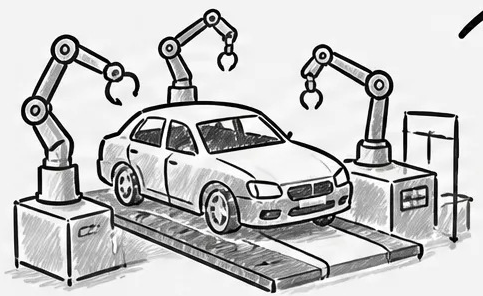
NODE
Visible within the node.

SERVER (TIGHTEST)
Visible only on the server.

Connection Pooling Mechanics

The Problem:

Instead of building a new car for every customer...

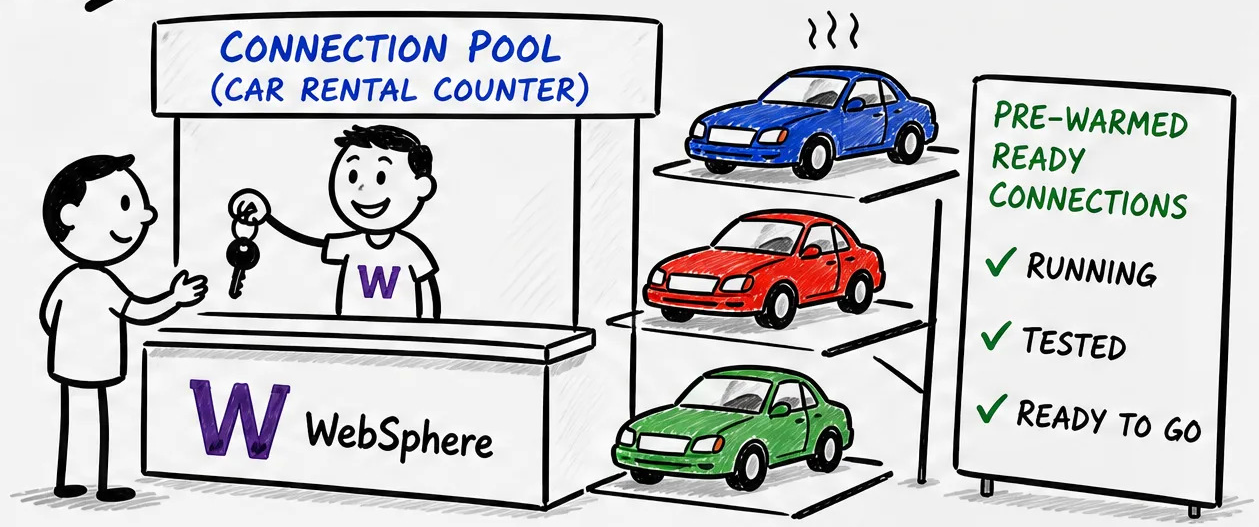


- ✗ Slow (Builds take time)
- ✗ Expensive (High CPU/Memory)
- ✗ Inefficient (Wasted resources)

Trading Memory for Speed

The Solution:

WebSphere hands them the keys to a pre-warmed, running car.





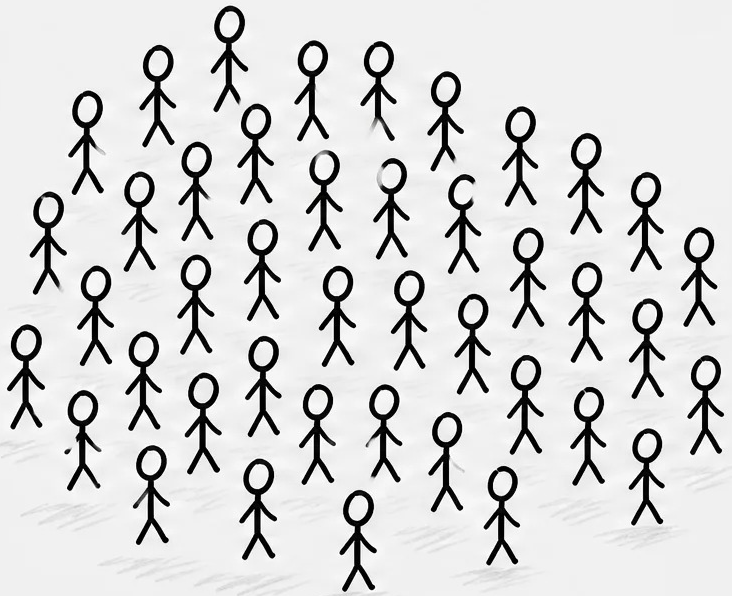
Connection Pooling trades Memory for Massive Performance Gains.

- ✓ Faster Response Times
- ✓ Lower CPU & Resource Usage
- ✓ Better Scalability
- ✓ Happier Users

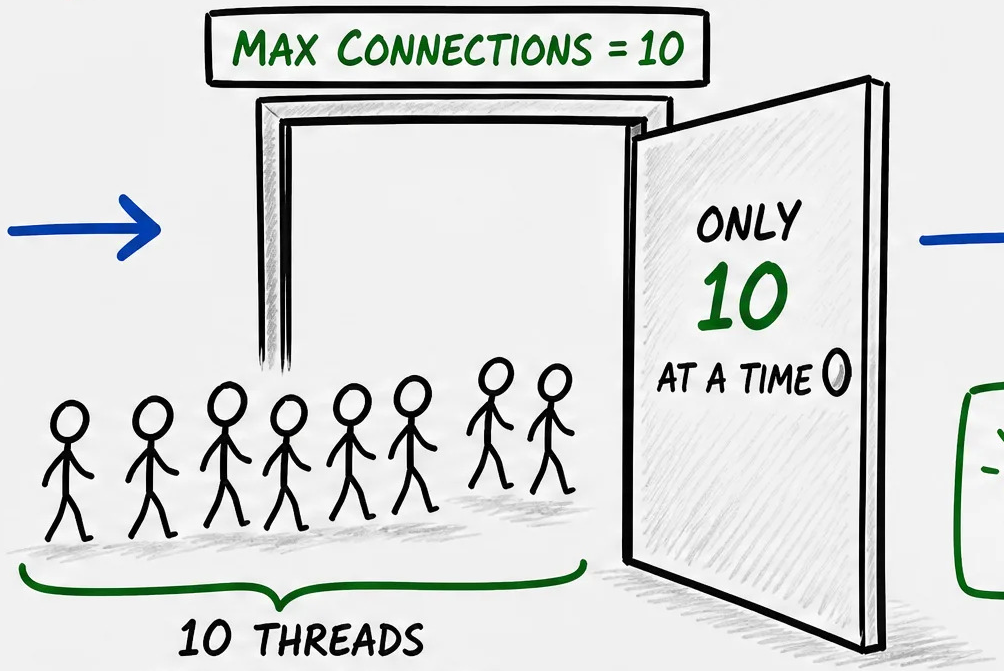
Tuning the Connection Pool

Avoiding Pool Starvation

Web Container Threads (50)

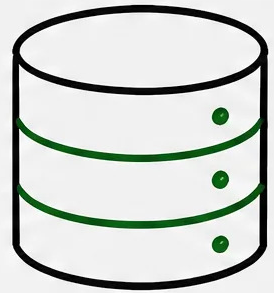


50 threads trying to get a connection



10 threads getting a connection

Database Connection Pool



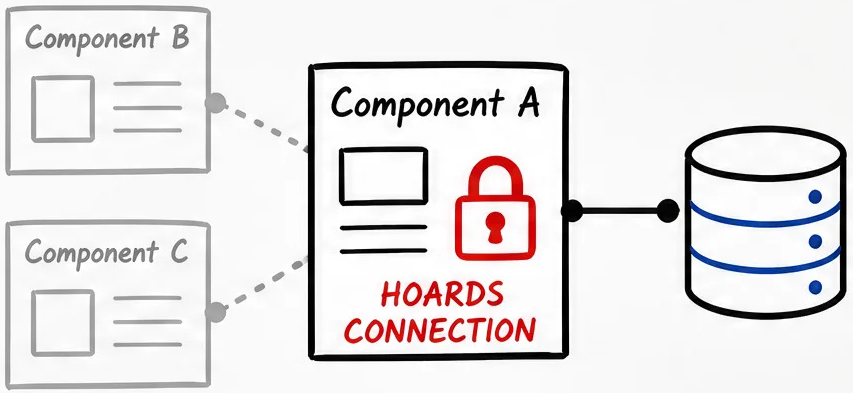


The Connection Pool Size must align with your Thread Pool Size.

Shareable vs. Unshareable Connections

UNSHAREABLE

One component hoards the connection

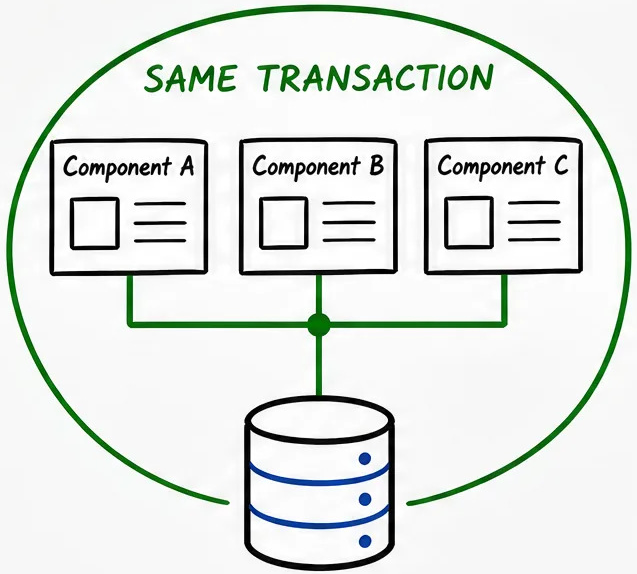


- ✗ Blocks other components
- ✗ Limits concurrency
- ✗ Wastes pool resources
- ✗ Increases wait and contention



SHAREABLE

Multiple components safely share one connection



- ✓ Enables concurrency
- ✓ Efficient resource use
- ✓ Lower wait times
- ✓ Scales better

Shareable connections conserve pool resources during complex transactions.

Local Transaction Containment (LTC)

LOOSE DATABASE WORK ITEMS

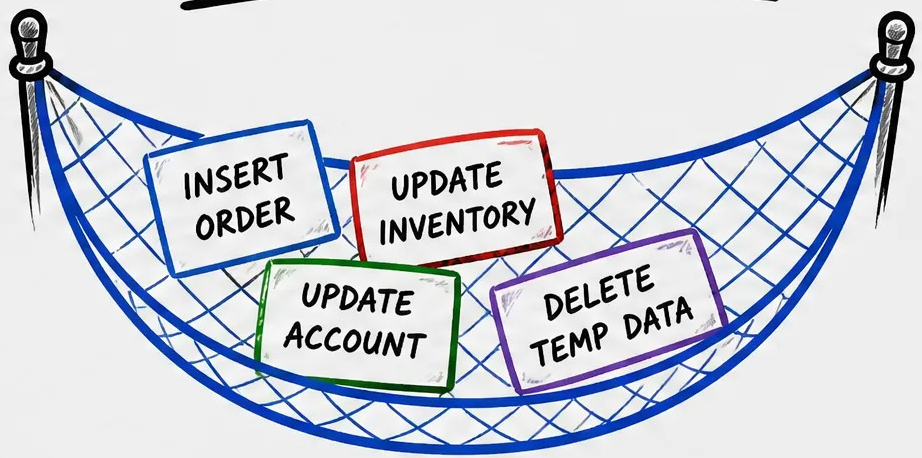
INSERT ORDER

UPDATE INVENTORY

UPDATE ACCOUNT

DELETE TEMP DATA

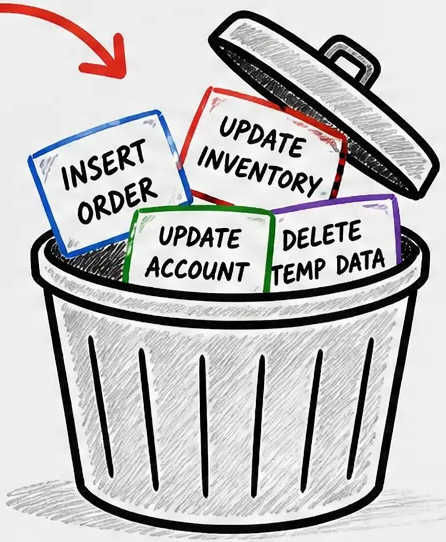
The Safety Net



LTC CONTAINS DATABASE WORK TO AN UNCOMMITTED, LOCAL CONTEXT.

CONTAINER CLOSES

All work items are automatically rolled back.



NOT COMMITTED.
NOT PERSISTED.
SAFELY DISCARDED.

The **LTC** acts as a **safety net** for unspecified transaction contexts.

Troubleshooting: Connection Leaks



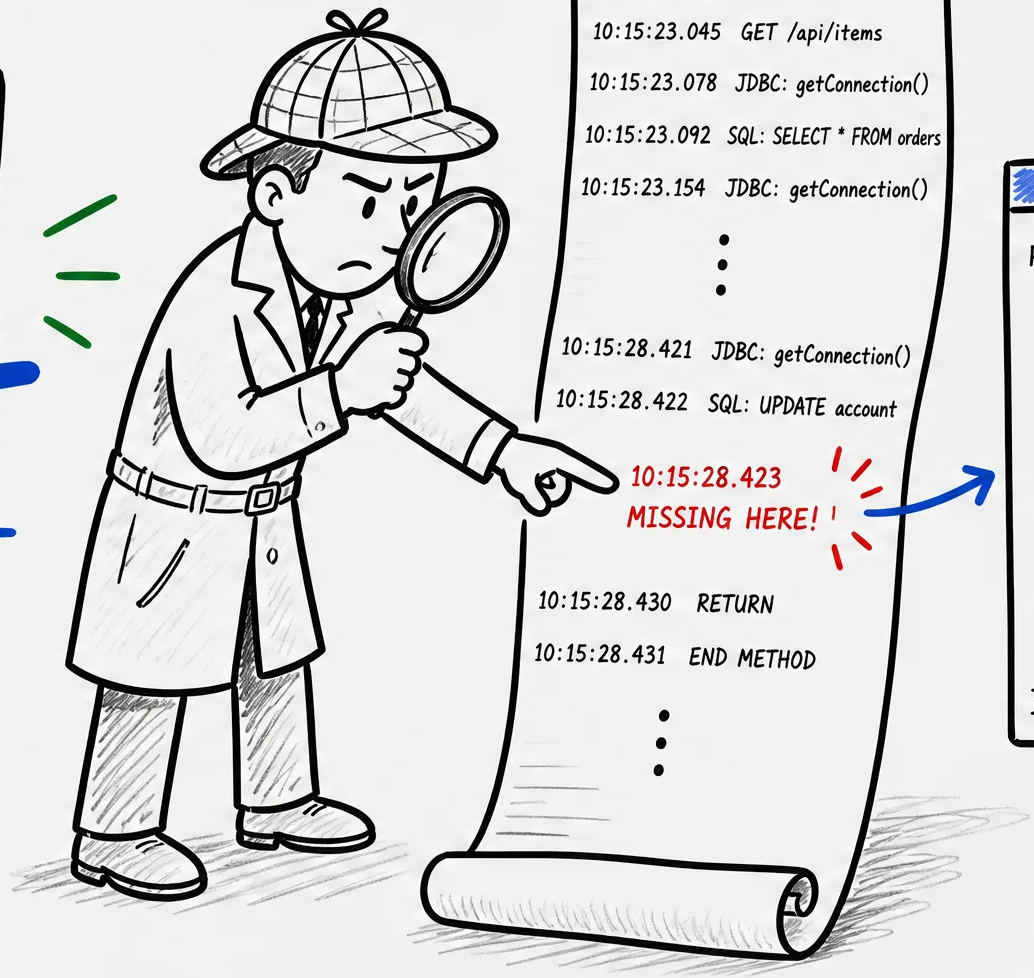
ConnectionWaitTimeoutException

The Missing .close()

A leaked connection is a **stolen resource.**

Diagnosing Leaks: ConnLeakLogic

Tracing the Culprit



BAD CODE IDENTIFIED!

```
public void updateAccount(int id)
    throws SQLExcpction {
    Connection conn =
      dataSource.getConnection();
    PreparedStatement ps =
      conn.prepareStatement(
        "UPDATE account SET
         balance = ? WHERE id = ?");
    ps.setInt(1, id);
    ps.executeUpdate();
    // MISSING: conn.close();
}
```

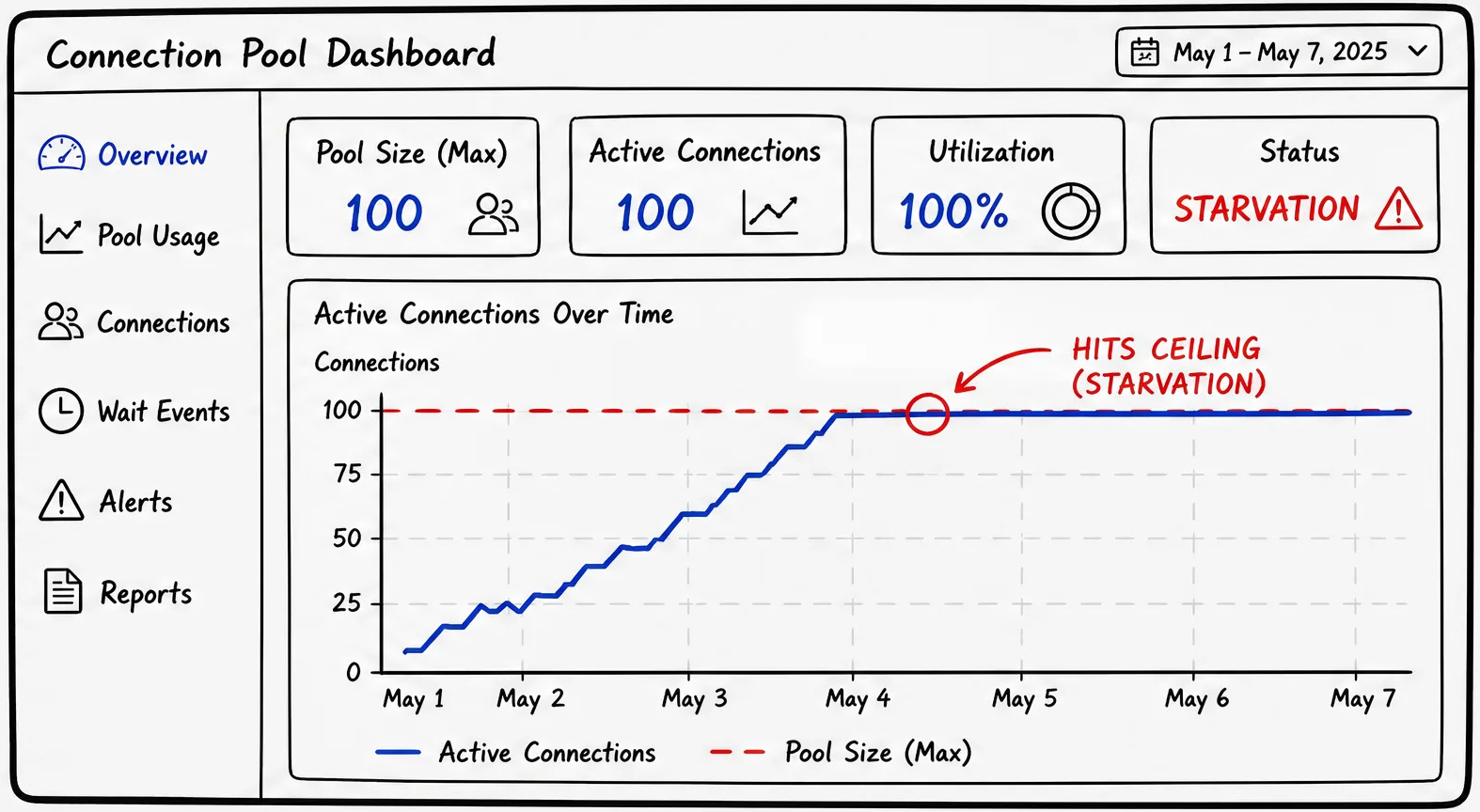


ConnLeakLogic points the finger directly at **bad code.**

IBM Database Connection Pool Analyzer

Visualizing the Pool

Visual analytics reveal patterns that raw logs hide.



Spot Issues Early

Reduce Starvation

Improve Reliability

Optimize Performance

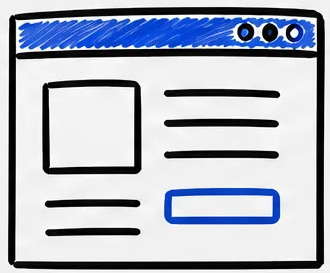
ProTech
protechtraining.com

Module 7 Summary

Dr. Lee

The Database Lifecycle

ALIAS



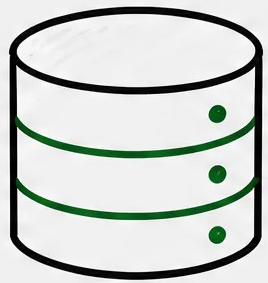
Logical name used by applications

PROVIDER



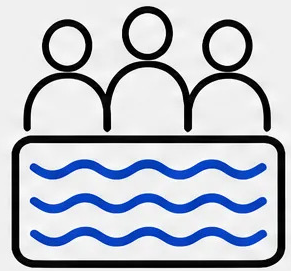
Manages connections, security, and resource use

DATA SOURCE



Configuration that defines how to connect

POOL



Reusable connections improve performance and scalability

TUNING / TRACE



Monitor, tune, and trace for optimal performance



WebSphere manages the connections so the application can focus on **the business.**